Environment: SDF Files

Flame Sensor:

Thermal camera (IR spectrum) for detecting fire heat.

RGB camera for visual flame detection.

IMU (accelerometer, gyroscope) for orientation.

Compass + GPS for global positioning.

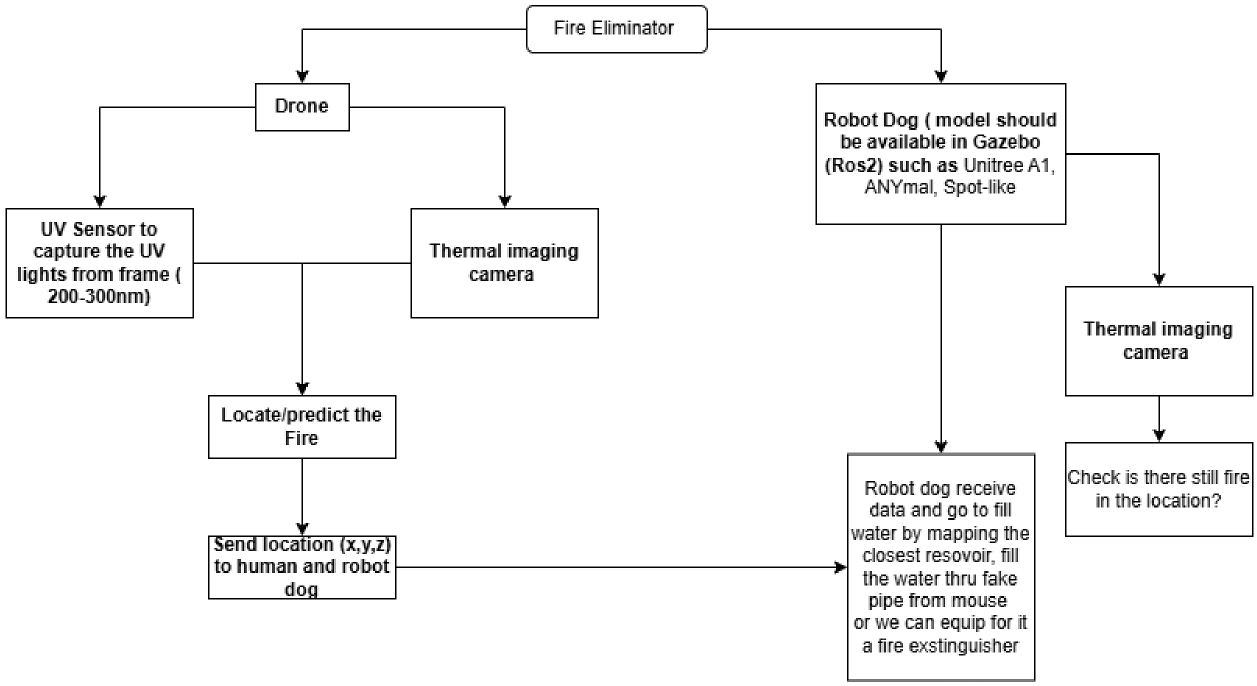
LIDAR (optional) for mapping obstacles and trees

Fire Detection Method:

- **Approach:**
  - Use **thermal imaging** to detect heat signatures.
  - RGB camera + AI (OpenCV or trained model) to detect flame colors and flicker.
  - STL or mesh models in simulation to represent fire objects for testing.

Simulation of Heat Signatures

- **Approach:**
  - Assign **temperature/intensity values** to fire objects in SDF/URDF.
  - Thermal camera plugin reads these values to produce a heat map.
  - Optional: Gaussian distribution for heat around fire center to simulate spread.


Fire Eliminator
Drone
Robot Dog (model should be available in Gazebo (Ros2) such as Unitree A1, ANYmal, Spot-like
UV Sensor to capture the UV lights from frame (200-300nm)
Thermal imaging camera
Thermal imaging camera
Locate/predict the Fire
Check is there still fire in the location?
Send location (x,y,z) to human and robot dog
Robot dog receive data and go to fill water by mapping the closest resovoir, fill the water thru fake pipe from mouse or we can equip for it a fire exstinguisher